BRYSON ROBERTSON

Professor, Mechanical Engineering | Director, Institute for Integrated Energy Systems
University of Victoria, British Columbia, Canada

RESEARCH & LEADERSHIP PROFILE

Marine renewable energy researcher and academic leader specializing in wave energy conversion, hydrodynamic modeling, resource characterization, experimental validation, and energy systems integration. Former Director of the Pacific Marine Energy Center (PMEC), with experience leading interdisciplinary research, open-source technology development, industry partnerships, and externally funded programs.

EDUCATION

Ph.D., Remote Measurement and Analysis of Shallow Water Wave Breaking Characteristics — Queen's University / University of Guelph, 2013

B.Eng., Mechanical Engineering — University of Victoria, 2005

ACADEMIC & PROFESSIONAL APPOINTMENTS

2026–present Professor, Mechanical Engineering, University of Victoria

2026–present Director, Institute for Integrated Energy Systems, University of Victoria

2018–2025 Associate Professor, Civil and Construction Engineering, Oregon State University

2018–2025 Director, Pacific Marine Energy Center, Oregon State University

2019–2024 Senior Scientist, Coastal Sciences Division, Pacific Northwest National Laboratory

2018–2023 Adjunct Professor, Institute for Integrated Energy Systems, University of Victoria

2013–2018 Faculty Researcher & Senior Research Engineer, University of Victoria

2006–2018 Founder & Director, OceanGybe Environmental

SELECTED RESEARCH CONTRIBUTIONS

- Led development of the open-source Laboratory Upgrade Point Absorber (LUPA), a configurable two-body WEC with physical prototype, numerical models, and curated validation data; recognized as a U.S. DOE Signature Project.
- Led SubWEC, a Navy-funded submerged WEC research platform combining experimental hardware, numerical models, and open validation datasets.
- Advanced experimental and numerical approaches for WEC design, PTO–mooring dynamics, uncertainty quantification, wave resource assessment, and hybrid simulation.
- Supported research linking marine energy to autonomous underwater vehicle docking and recharging, offshore wind, and resilient electricity systems.

SELECTED FUNDED RESEARCH (PI / CO-PI)

- DOE WPTO, National Marine Energy Centers (2024–2029): $14.7M total project budget.
- DOE WPTO, Applied Research and Development to Support Open-Water Testing at PacWave (2022–2025): $4.125M.
- NAVFAC Marine Energy FY21 (2021–2024): $2.4M; FY24 (2024–2028): $1.77M, plus $1.083M supplement.
- DOE WPTO, WEC engineering co-design for open-source performance improvement and LCOE reduction (2026–2029): $500K.
- DOE WPTO, wave-to-wire prototype development for autonomous underwater vehicle docking and recharging (2026–2029): $1M.

SELECTED PEER-REVIEWED PUBLICATIONS

- Okushemiya, D., Robertson, B., & Rusch, C. (2026). Performance and feasibility assessment of a wave energy converter with underwater vehicle docking and charging. Applied Energy, 409, 127453.
- Beringer, C., Bosma, B., & Robertson, B. (2026). Detailed uncertainty analysis and nonlinear effects of wave energy converter experimental testing in regular waves. Ocean Engineering, 348, 123024.
- Okushemiya, D., Rusch, C., Robertson, B., & Zhang, Z. (2026). Model validation and improved PTO modeling of a field-deployed wave energy converter with tethered heave plate. Applied Ocean Research, 166, 104921.
- Beringer, C., Bosma, B., & Robertson, B. (2025). A numerical and experimental study of wave energy converter number of bodies and degrees of freedom. Renewable Energy, 125089.
- Roach, A., Meek, M., Ali, R., DuPont, B., & Robertson, B. (2025). A state-of-the-art review of submerged wave energy converters. Renewable and Sustainable Energy Reviews, 222, 115901.
- Ali, R., Meek, M., & Robertson, B. (2025). Submerged wave energy converter dynamic impacts from mooring system configurations. Renewable Energy, 243.
- Chen, M., et al. (2024). A unified simulation framework for wave energy powered underwater vehicle docking and charging. Applied Energy, 361, 122877.

TEACHING & RESEARCH MENTORSHIP

Courses taught: Ocean Engineering (OSU CE 411/511); Hydraulic Engineering (OSU CE 313); Hydraulics (OSU CEM 311); Multidisciplinary Capstone (OSU ENGR 415/416); Introduction to Ocean Engineering (UVic MECH 446).

Supervised graduate research across WEC hydrodynamics, wave resource uncertainty, submerged systems, marine energy integration, and hybrid simulation; mentored postdoctoral researchers and multidisciplinary student teams.

Organized and delivered the Wave Energy Converter Design and Control Workshop (2024) and Navy Waves Workshop (2026); faculty advisor for DOE Marine Energy Collegiate Competition teams (2021–2025).

SELECTED PROFESSIONAL LEADERSHIP & SERVICE

- Associate Editor, Maritime Engineering – Proceedings of the Institution of Civil Engineers (2022–present).
- Guest Editor, Applied Ocean Research special issue on metocean data and numerical innovations (2025–2026).
- Member, IEC TC-114 shadow committee on marine energy standards (2014–present).
- Member, National Hydropower Association Marine Energy Council Board (2022–2025); DOE TEAMER Management Board (2019–2025).
- Invited testimony on renewable energy before Oregon House of Representatives (2025) and U.S. House of Representatives (2019).

SELECTED RECOGNITION

Faculty advisor to DOE Marine Energy Collegiate Competition teams receiving 2nd place overall (2022), 3rd place overall (2023), and multiple Build and Test / Outreach awards (2021–2023).

Source: Bryson Robertson, Curriculum Vitae, May 4, 2026. Selected highlights; full publication and funding records available in the original dossier.